

QUA'TARIOUS A. BOWMAN

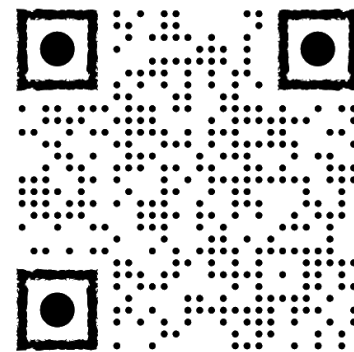
B.S. MECHANICAL ENGINEERING • U.S. CITIZEN (CLEARANCE ELIGIBLE)

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Portfolio: resume-pink-three.vercel.app



portfolio QR this way!
*its pretty awesome



EDUCATION

University of Virginia

B.S. Mechanical Engineering

Expected May 2028

Practical Core GPA: 3.53 / 4.00

ME Major GPA: 2.79 / 4.00

PRACTICAL COURSEWORK

- **Mechanics Lab (A+):** Calibrated transducers/gauges, strain testing, & stress analysis.
- **CAD (A-):** Parametric 3D modeling, mechanical assemblies, & drafting standards.
- **Strength of Materials (B+):** Beam deflection, stress transform, & failure evaluation.
- **Physics Workshop (A-/B+):** Applied mechanics & hardware validation.

TECHNICAL ARSENAL

CAD & MODELING

- SolidWorks & Fusion 360
- Parametric Part Modeling
- Complex Mechanical Assemblies

FABRICATION & HANDS-ON

- Micro Soldering (IPC Standards)
- FDM 3D Printing (ABS/PLA)
- Structural Carpentry
- Mixed-Material Prototypes

HARDWARE SYSTEMS

- LEV Powertrains & Geometry
- ESC / Flight Controller Stacks
- High-Current Wire Harnesses

RAPID PROTOTYPING TOOLS

- Python Automation Scripting
- React, Tailwind CSS, Git

ENGINEERING & HARDWARE PORTFOLIO

Potential Powered Mini Kart | Lead CAD & Prototyping

Spring 2026

- Co-designed and fabricated a dual-propulsion vehicle with a two-spring potential energy drivetrain, 10F supercapacitor-driven propeller, and 2.5:1 ratio gear train within tight 12"x8"x4" physical bounds.
- Iterated 4+ chassis geometries in SolidWorks to achieve a lightweight 20.62g ABS frame; verified kinematics against empirical performance across a 60-ft competitive track.

Doordasher™ (Custom 32mph LEV) | Systems Integrator

Summer 2026

- Integrated a 1000W rear-wheel drive e-bike powertrain under a lean design-to-production budget (<\$600); engineered top-mounted battery enclosure to resolve spatial limits in main frame triangle.
- Identified and corrected factory frame manufacturing defects, embedded concealed security tracking hardware, and validated reliability over 200+ miles of high-load commercial operation (DoorDash).

Custom 5-Inch FPV | Electronics Tech

Spring 2026

- Executed micro-scale precision soldering to integrate 4-in-1 ESC and Flight Controller stack on a high-density carbon-fiber frame.
- Performed continuity testing, diagnostic noise isolation, and custom wire harness routing; logged extensive manual/simulated flight hours evaluating real-time analog/digital telemetry.

Virtual Reality Jarvis Station | Integration Designer

Summer 2026

- Engineered a overhead structural cable harness and continuous power delivery system incorporating vibration-dampening geometry to reduce mechanical wear on desktop components during extended VR use.

Personal Portfolio Web Application | Developer

Summer 2026

- Architected an interactive engineering showcase using React, Tailwind CSS, and Framer Motion with custom JSON admin workflows and real-time telemetry display.